

Reading the Fine Print

LLM-Scored Pharma 8-K Filings as a Sector-Neutral Equity Signal

Quant note · Gator Quant Hacks 2026, Track 03: Systematic Trading · October 4, 2026

Code and data scripts: github.com/noahhassan2026/gator-quant-hacks-pharma

1. Summary

This note studies whether the text of material 8-K filings by U.S. pharmaceutical and biotechnology companies predicts subsequent returns relative to the sector. For each filing, an open-weight large language model (gpt-oss-20b) scores sentiment, surprise, and materiality. A gradient-boosted classifier combines these scores with price and liquidity features to estimate the probability that the stock outperforms the SPDR S&P Pharmaceuticals ETF (XPH) over the following five trading days, and a second model selects the exit rule. Every position is hedged with XPH, so returns reflect the filing's information rather than sector movements.

Test	Period	Trades	Avg net trade	Return	Sharpe	Max DD
Walk-forward (in-sample)	2020-2024	3,850	+1.5%	n.c.	n.c.	n.c.
Held-back test	2025-02-18 to 2026-10-02	1,095	+0.91%	+60.8%	2.12	-10.3%
XPH buy-and-hold	same	n/a	n/a	+41.2%	1.02	-18.7%

All results are net of 10 bps per side and estimated short-borrow fees; n.c. denotes a statistic not computed before submission (Section 5). **Two caveats are material.** First, we evaluated the held-back period three times and chose the stop rule after the first evaluation, so we treat it as a validation set rather than a locked test. Second, the result depends on checking stops at the daily close: when stops are checked every minute, the same signals earn approximately zero (-0.6%). Both results are reported.

2. Economic hypothesis: who takes the other side?

Hypothesis. We expect small and mid-cap pharmaceutical stocks to **underperform XPH over 3 to 10 trading days** following certain material 8-Ks, such as dilutive financings, ambiguous trial readouts, and listing-compliance notices. **The counterparty** is predominantly retail and thematic investors who respond to headlines and press releases rather than to the filing itself, and whose expectations remain optimistic. When short selling is costly, prices reflect the views of the most optimistic holders (Diether et al., 2002; Miller, 1977). **The edge persists** because of one structural constraint and one behavioral bias. Borrow fees, limited borrow availability, and squeeze risk deter arbitrage capital from small biotechnology stocks, and filings are long and technical, so few market participants read them closely (Loughran & Ritter, 1995; Tetlock, 2007). **Testable prediction:** net of costs and borrow fees, model-selected trades outperform a naive rule that shorts every filing. **Failure condition:** the walk-forward average trade is not positive in most years, or the model adds no value over shorting every filing.

History of the hypothesis. Our original hypothesis, committed to the repository on 2026-10-03 at 18:48 ET before any backtest, concerned Hatch-Waxman patent litigation; an LLM would read judge assignments and court rulings, and positions would be hedged with XPH. Because we could not assemble sufficient historical court data in the time available, we broadened the same idea from court dockets to SEC filings. The universe, hedge, LLM use, and execution plan are unchanged. We disclose that we wrote this revision after observing early results.

3. Data and universe

Source	What we use	Detail
SEC EDGAR (free)	8-K list, item codes, acceptance time, industry code	SIC 2834, 2835, 2836, 8731; items 1.01, 1.02, 2.01, 2.02, 2.05, 2.06, 3.01, 5.02, 7.01, 8.01
Massive (Polygon)	Split-adjusted daily bars; 1-minute bars	Minute bars for entries; XPH as hedge and benchmark
gpt-oss-20b (UF NaviGator)	Text scores from each filing's first 6,000 characters	Structured JSON output; 5 filings per request

Sample. Between 2016 and 2024, pharmaceutical companies filed 39,650 material 8-Ks, of which 16,390 were tradable (share price of at least \$5 and average daily dollar volume of at least \$1M). Between 2025-02-15 and

2026-10-02, there were 10,674 filings, of which 4,174 were tradable. **Sample splits.** Training and validation use 2016-2024. The held-back test begins on 2025-02-15, the first day after the outcome window of the last training label closes. The track guidelines call for the most recent two years; our held-back period covers approximately 1.6 years because October 2024 to February 2025 overlaps the training labels. **Data issues.** Prices are split-adjusted. Where minute bars are missing, the daily close is used; gaps are never filled with later data. Tickers are mapped using the current SEC ticker list, so some delisted companies are absent. This survivorship bias probably *understates* the performance of a short-biased strategy, since failed biotechnology companies would have been profitable shorts, but we have not quantified it. The LLM's training data extends to approximately mid-2024, so the model may recall some 2016-2024 outcomes. Filings in the held-back test postdate its training cutoff and are not subject to this leakage.

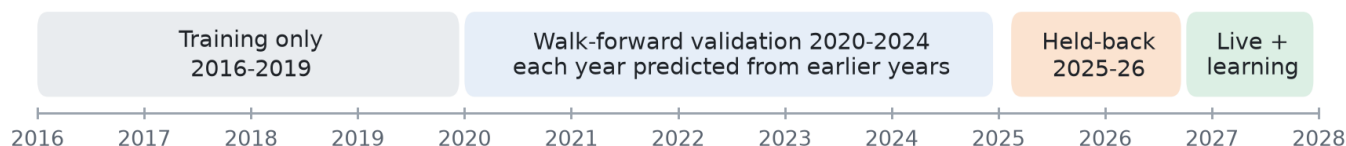


Figure 1. Use of each data period. Apart from the stop rule (Section 5), no setting was chosen using data after 2024.

4. Methodology

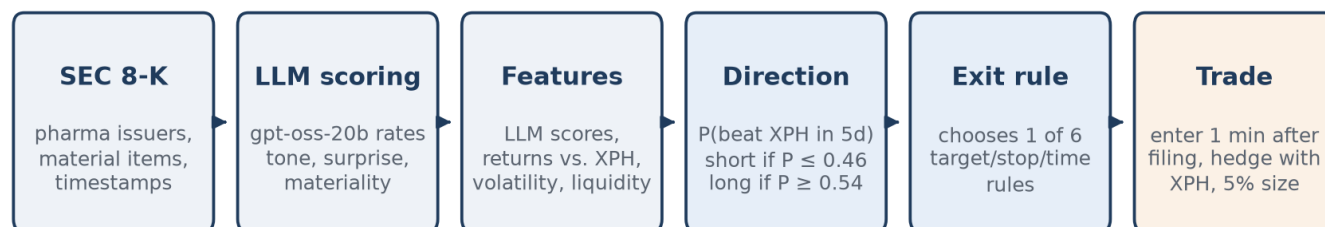


Figure 2. From filing to trade. All inputs are available before the entry time.

- **No look-ahead.** A signal exists only once the SEC has accepted the filing. Positions are entered at the first regular-hours minute at least one minute after acceptance; filings accepted outside market hours are entered at the next open.
- **Features.** The LLM's sentiment, surprise, materiality, upcoming-catalyst, and event-type outputs; 5- and 20-day returns relative to XPH; 20-day volatility; dollar volume; share price; an after-hours indicator; days since the company's previous filing; and 8-K item codes.
- **Models.** Both models are gradient-boosted decision trees implemented in scikit-learn (Pedregosa et al., 2011). The direction model estimates the probability that the stock outperforms XPH over five trading days; the strategy goes long when this probability is at least 0.54 and short when it is at most 0.46. The exit model selects one of six rules, specified as target/stop/maximum holding days: 5%/3%/3, 8%/5%/10, 15%/8%/30, 25%/12%/30, none/15%/20, and a volatility-scaled rule.
- **Model selection.** We use walk-forward validation over 2020-2024, in which each year is predicted by a model trained only on earlier years, with a 45-day gap between training and test data. We compared 37 model settings and 12 probability cutoffs (444 combinations). A combination was eligible only if it produced at least 150 trades and a positive average trade in at least four of five years; among eligible combinations, we selected the one with the highest t-statistic. We refit the final model on all data from 2016 to 2024.
- **Costs.** A cost of 10 bps per side covers the half-spread and slippage on \$5k orders; Webull charges no commission on U.S. equities apart from small regulatory fees. Borrow fees follow Webull's formula (position value $\times$ annual rate / 360 per day), with the annual rate estimated from liquidity: 0.5% for daily dollar volume of at least \$100M, 2% for at least \$20M, 8% for at least \$5M, and 25% otherwise (0.5% for XPH).
- **Portfolio.** The portfolio starts with \$100k and allocates 5% of equity per trade, with at most 20 open positions, one position per ticker, and gross exposure capped at 100%. Stops and targets are evaluated at each

day's close, consistent with the construction of the training labels.

5. Results

Sources of the edge. Under identical exits and costs, the model's trades averaged +1.36%, compared with +0.12% for shorting every tradable filing, which indicates that the model adds genuine selection beyond a simple short bias. Clinical-trial results (+3.2% across 168 trades) and mergers and acquisitions (+6.8% across 17 trades) contributed most. Trades following FDA decisions (-0.6%) and patent litigation (-1.6% across 21 trades) lost money, which is evidence against our original litigation hypothesis. Because stops are evaluated only at the close, losses can gap beyond the 3% stop: the average loss was -8.2% and the average gain +9.1%.

Net of costs	Walk-forward 2020-24	Held-back test	Held-back, costs doubled (est.)
Average/median trade	+1.5% / n.c.	+0.91% / +1.38%	about +0.71%
Win rate	57.0%	52.7%	n.c.
Annualized return/volatility	n.c.	+34.0% / 15.4%	about +25% / 15.4%
Sharpe / Sortino	n.c.	2.12 / 2.56	about 1.7
Turnover (x equity per year)	about 77x	about 68x	about 68x

The doubled-cost column assumes that each additional 20 bps round trip on a 5% position costs 0.01% of equity. Turnover counts both sides of each trade. Statistics marked n.c. are produced by `report_metrics.py` in the repository, which was not run before the submission deadline.

Held-back test: strategy vs. XPH, February 2025 to October 2026

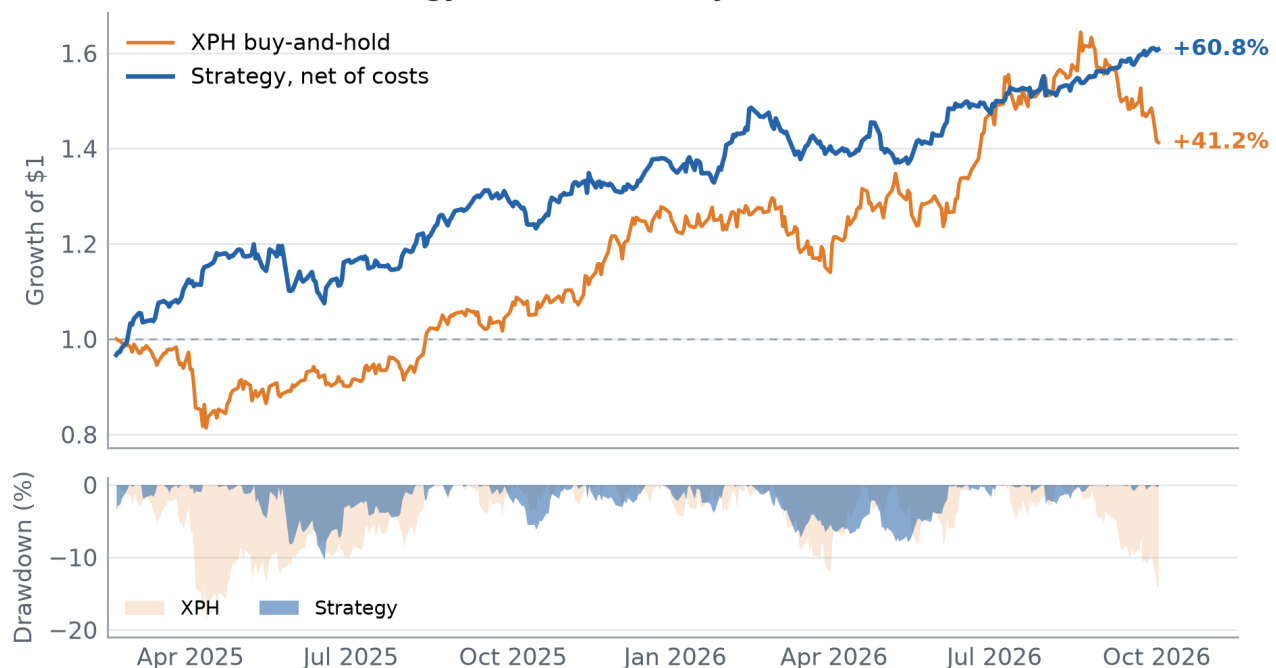


Figure 3. Held-back test, daily: growth of \$1 for the strategy (net of all costs) and for XPH buy-and-hold (top), and the strategy's drawdown (bottom). The strategy ends at \$1.61 versus \$1.41 for XPH, with about half of XPH's worst drawdown (-10.3% vs. -18.7%) and almost no correlation (-0.10).

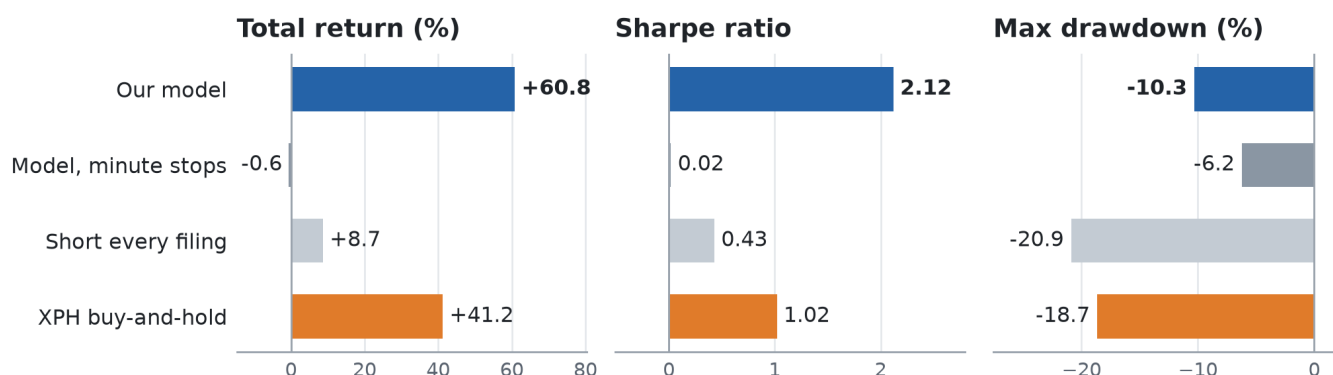


Figure 4. Performance over the same period relative to XPH and two alternatives. All figures are net of costs; the model and the short-every-filing rule use identical exits.

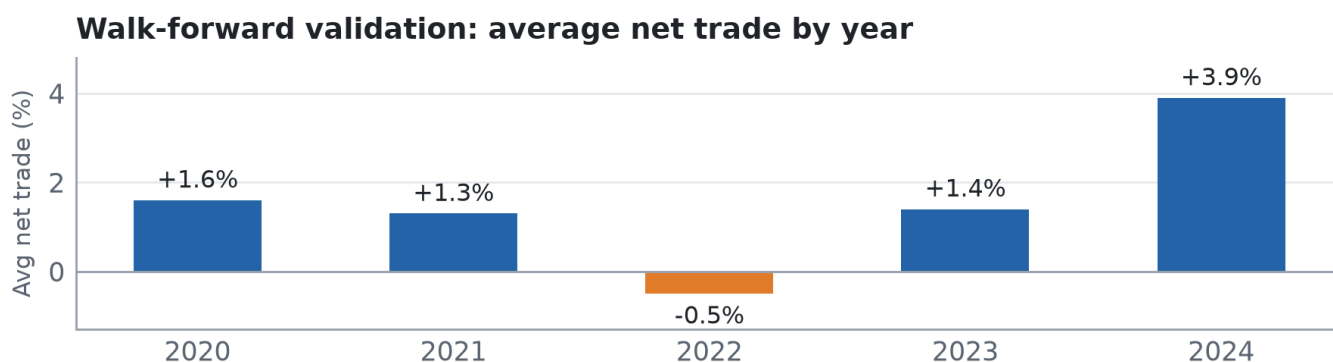


Figure 5. Walk-forward average net trade by year. 2022, a bear market for biotechnology, was the only losing year.

Variants tested and failures. We tested approximately 462 variants in total: 444 walk-forward combinations, an earlier model with 12 cutoffs tuned on 2023-2024, six held-back runs of the strategy, and four baseline runs. The first held-back run (earlier model, stops checked every minute) returned +23.1% with a Sharpe ratio of 1.44, but 382 of 692 trades were stopped out. We then observed that the training labels evaluated stops only at the close, aligned the backtest accordingly, and obtained +103.9% with a Sharpe ratio of 3.02. We regard that figure as suspect, since a Sharpe ratio above 3 usually indicates an error. We subsequently retuned the model using only 2020-2024 data, added borrow fees, and ran the held-back test a third time to obtain the headline result. With minute-level stops, the tuned model returns -0.6%. Confidence-based sizing never outperformed fixed sizing. **Multiple testing.** The selected setting's walk-forward t-statistic is 6.8, compared with a Bonferroni threshold of 3.7 for 462 tests and the hurdle of approximately 3.0 proposed by Harvey et al. (2016); because trades overlap in time, the effective t-statistic is lower. The repository implements the Deflated Sharpe Ratio (Bailey & López de Prado, 2014), but we did not compute it in time, so we do not report it.

6. Risk management

Risk	Control in the backtest	Added rule for live trading
Single-name loss	5% per name; 3% stop (0.15% of equity if it fills)	Hard intraday stop at -20% per position to cap squeezes
Concentration	At most 20 names, one per ticker; gross at most 100% (average 27% used)	At most 20% of equity in names under \$5M daily volume
Sector / market	Every leg hedged 1:1 with XPH (realized beta to XPH: -0.06)	Monitor beta to SPY and XPH each month
Drawdown	None	Halve size at -10%; stop new trades at -15%; restore after a new high
Borrow / recall	Fees estimated from liquidity (0.5% to 25%)	Skip hard-to-borrow names or rates above 25% on Webull
Model decay	Fixed model	Champion/challenger retraining (Section 8); halt if the 3-month live average trade turns negative

The principal risk is an adverse price gap on a short position: a single biotechnology stock can rise 50% or more on acquisition or trial news, which would cost at least 2.5% of equity at 5% sizing. **Factor exposure.** A

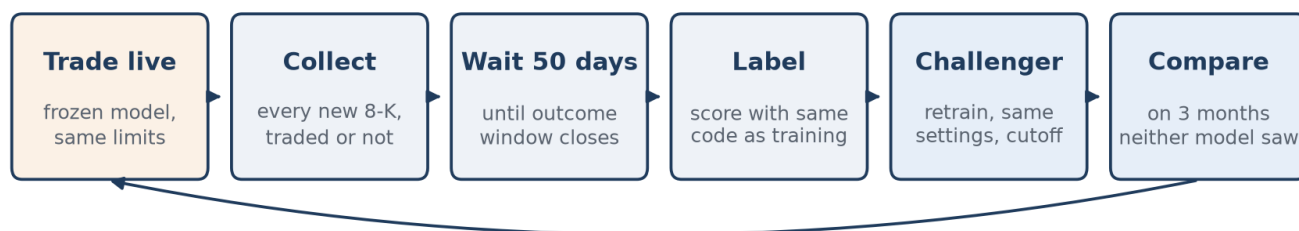
regression of daily held-back returns on XPH yields a beta of -0.06 (correlation -0.10, R^2 of 1%) and an annualized alpha of 34% ($t = 2.8$), indicating that the returns are not driven by sector exposure. Exposures to the market (SPY), size, and momentum have not yet been measured; an all-short portfolio of small-cap stocks may load on size and momentum. The live-trading rules in the table above are proposals and have not been backtested.

7. Liquidity and capacity

Every traded stock has an average daily dollar volume of at least \$1M. With \$100k of capital, each order is \$5k, which is at most 0.5% of daily volume in the least liquid names and considerably less in most. Participation rises in proportion to capital: at \$1M, orders in the least liquid names reach 5% of daily volume. At that level, square-root market impact (Almgren et al., 2005), approximately 0.5 to $1 \times \text{daily volatility} \times \text{the square root of participation}$, or roughly 50-100 bps for a stock with 4% daily volatility, would exceed our cost assumption and absorb most of the +0.9% average edge. Doubling costs to 20 bps per side still leaves an average trade of approximately +0.71%. **We estimate capacity at approximately \$1-5M** as currently designed, and higher if each order were capped at 1% of the stock's daily volume, which would exclude or reduce positions in the least liquid names. For the smallest stocks, short availability is likely to become binding before market impact.

8. Live trading and continuous learning

The system is designed to continue learning once it is deployed. A weekly job (`live_learning.py`) downloads every new pharmaceutical 8-K, waits 50 calendar days until the longest exit window for each filing has closed, and then labels and scores the filing with the same code used in training. It learns from **every** filing, not only those it traded, so it also learns from the signals it declined. It then trains a challenger model with the same settings and cutoff; settings are never re-tuned on live data, which guards against fitting to noise. The challenger replaces the current model only if it performs at least as well over the most recent three months, which neither model has seen. Every replaced model is archived to allow rollback, and every run is logged. None of the reported results use this loop; the first deployment step is paper trading on Webull with the frozen model.



Promote only if the challenger performs at least as well; the previous model is archived for rollback. Runs weekly.
Figure 6. The weekly learning loop. A new model is promoted only on out-of-sample evidence, and rollback is always possible.

9. Limitations and next steps

- **Reuse of the held-back period and dependence on the stop rule.** The minute-stop version, which is fully tradable, earns approximately zero. The close-check version assumes that the exit fills at the same close that triggered it; in practice, we would evaluate stops at 15:55 and submit a market-on-close order. As a next step, we will test exits at the following open and evaluate the strategy once on fresh data.
- **Data.** Survivorship bias from the current ticker list, estimated rather than observed borrow rates, possible LLM recall of 2016-2024 events, and LLM scores that vary slightly between runs.
- **Statistics not yet computed.** The Deflated Sharpe Ratio, the walk-forward daily Sharpe ratio and drawdown, exact doubled-cost results, the distribution of trade size relative to daily volume, and betas to SPY, size, and momentum. All are implemented in `report_metrics.py`.

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